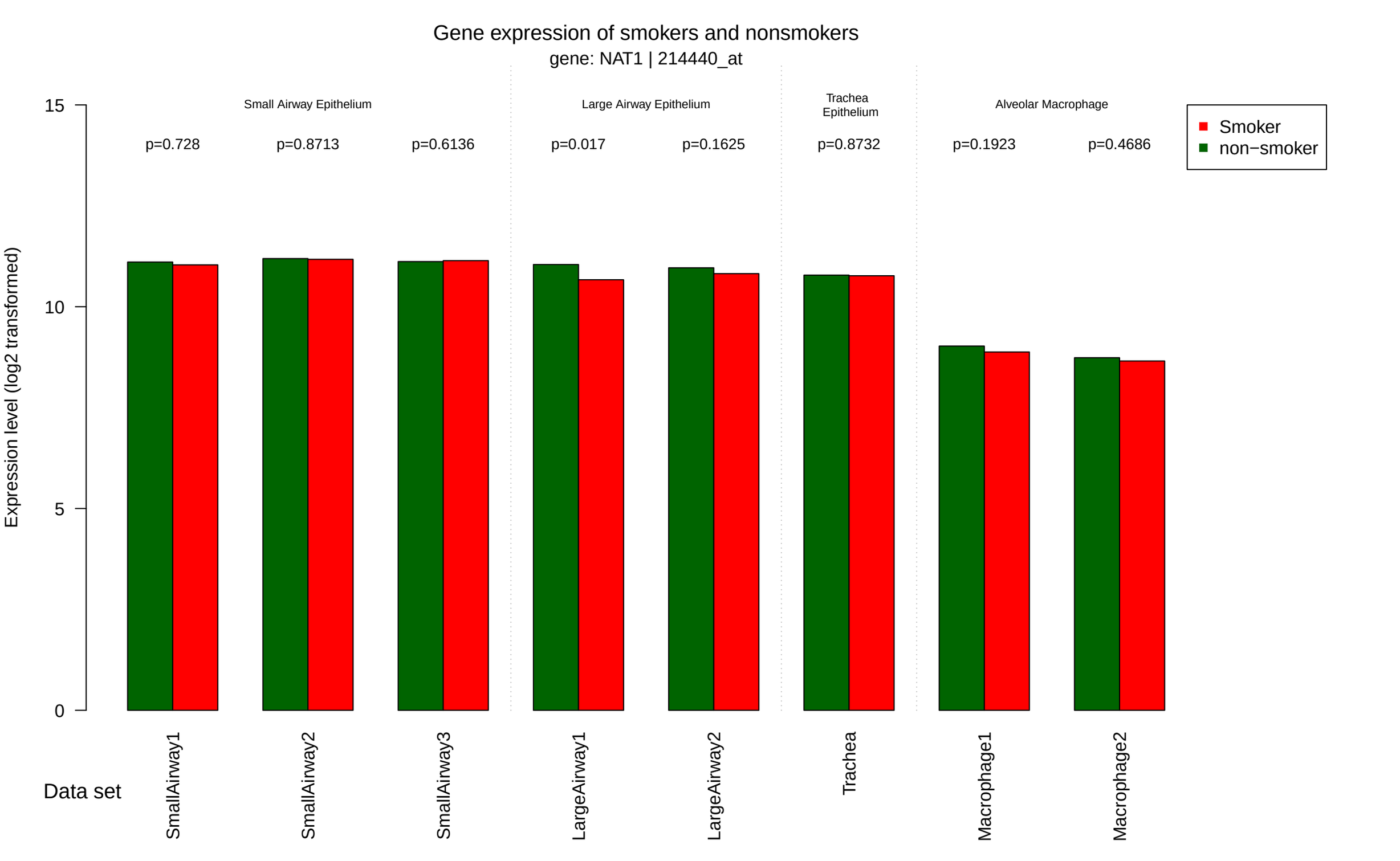
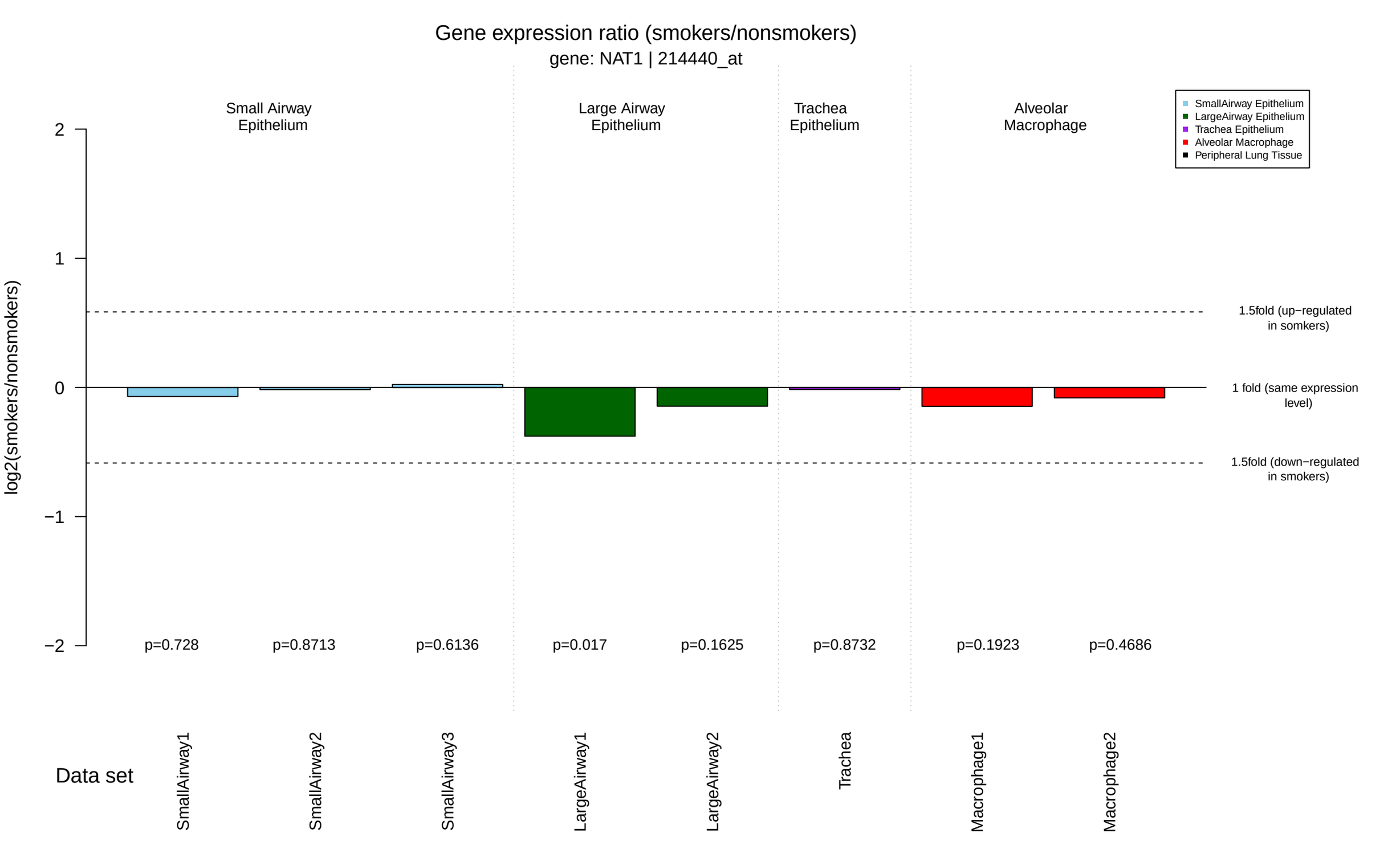
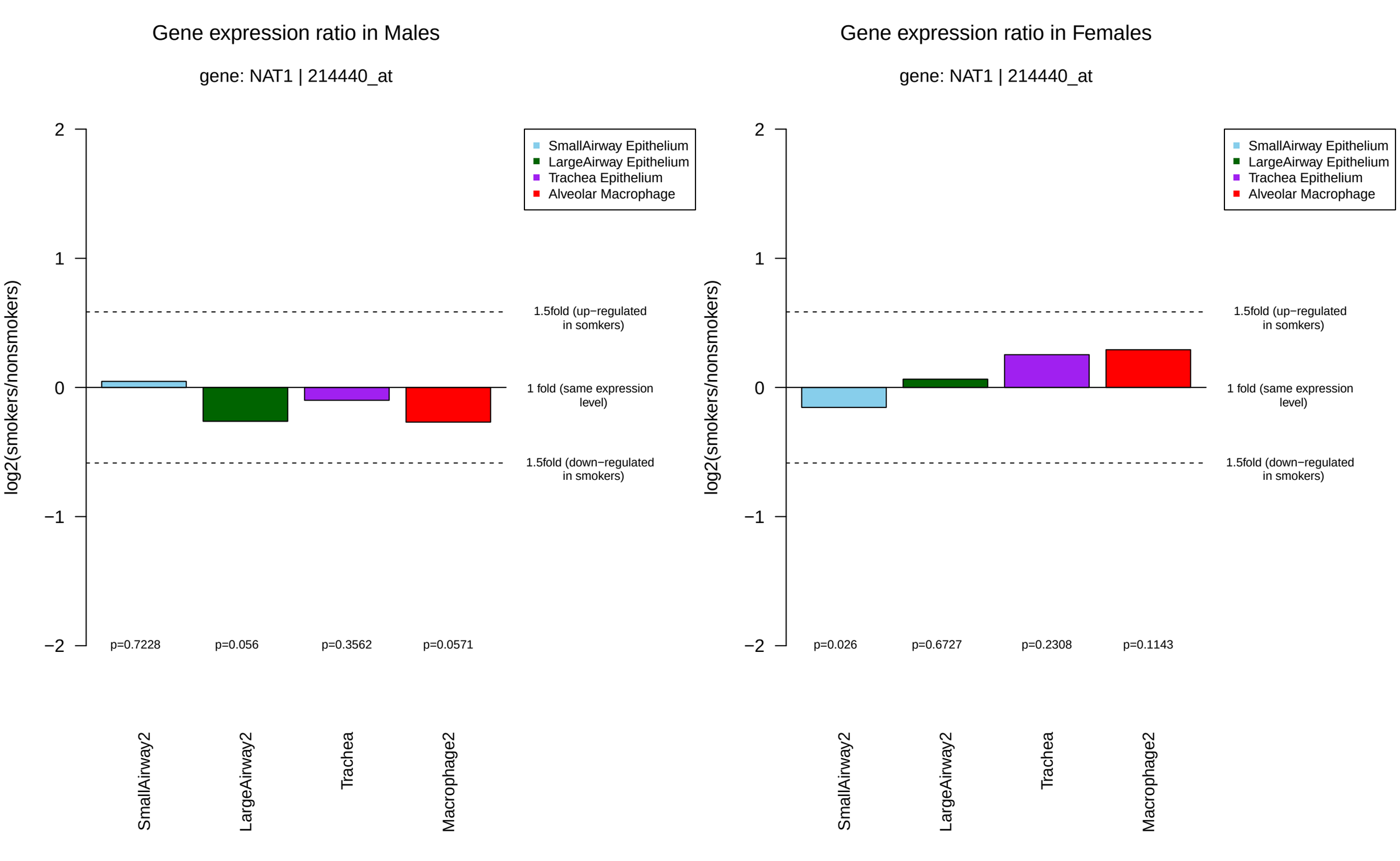
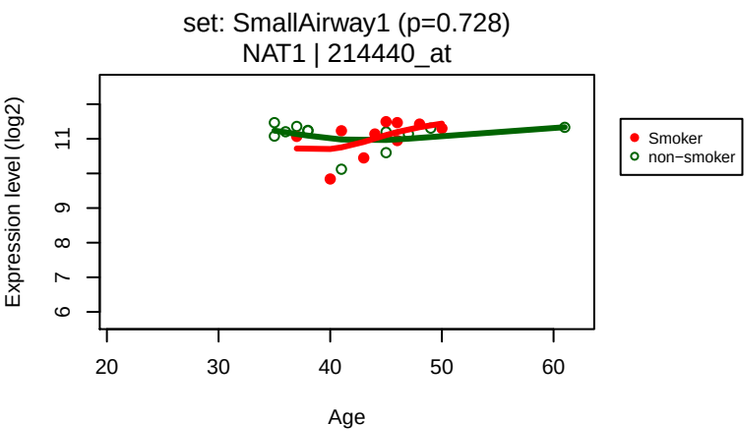
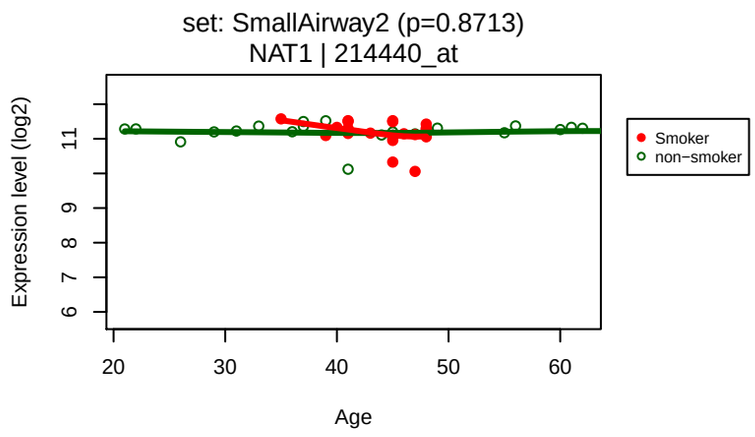
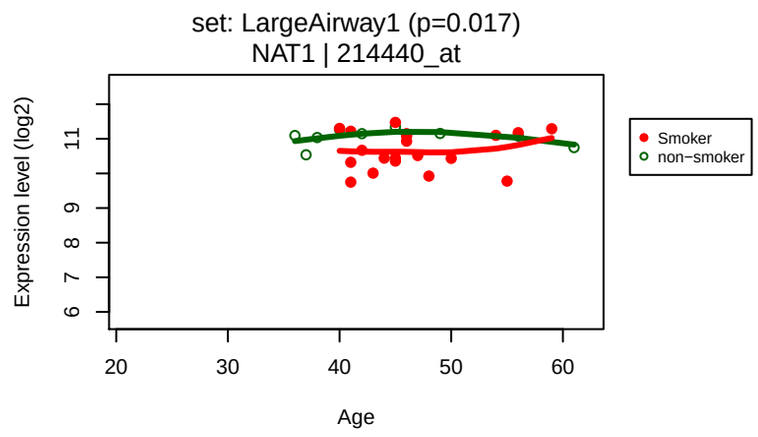

Gene expression of smokers and nonsmokers
gene: NAT1 | 214440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.728
p=0.8713
p=0.6136
p=0.017
p=0.1625
p=0.8732
p=0.1923
p=0.4686
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

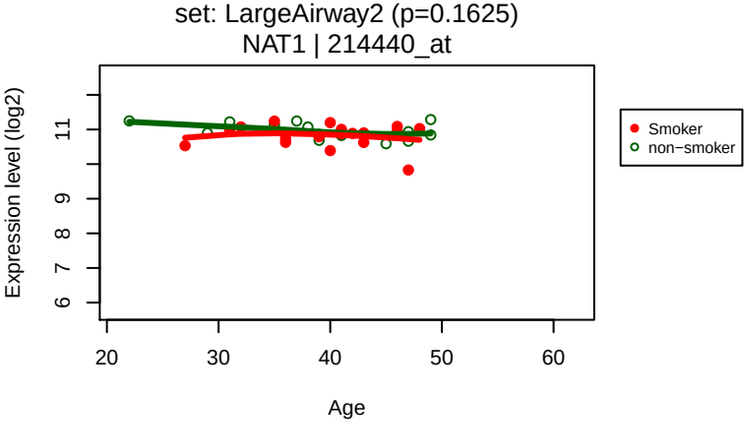

Gene expression ratio (smokers/nonsmokers)
gene: NAT1 | 214440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.728
p=0.8713
p=0.6136
p=0.017
p=0.1625
p=0.8732
p=0.1923
p=0.4686
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

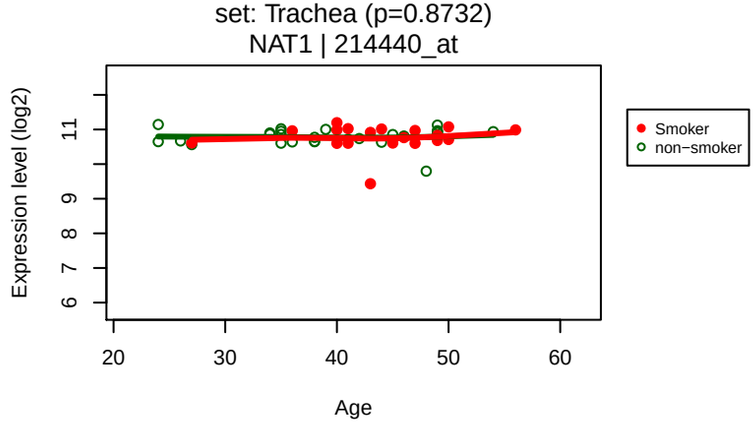

Gene expression ratio in Males
gene: NAT1 | 214440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7228
p=0.056
p=0.3562
p=0.0571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NAT1 | 214440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.026
p=0.6727
p=0.2308
p=0.1143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

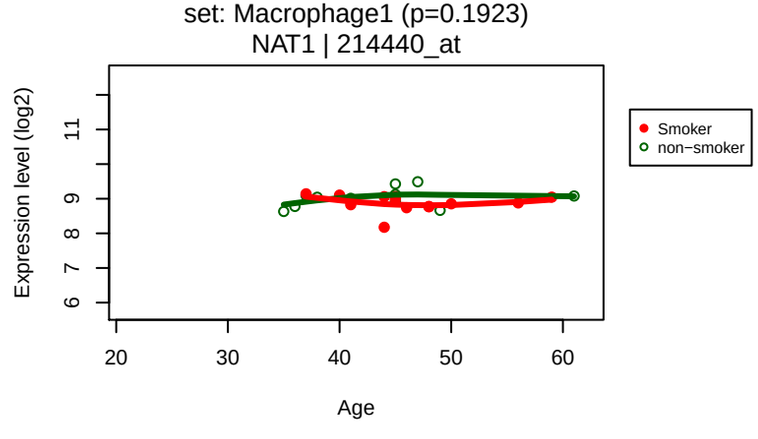

set: SmallAirway1 (p=0.728)
NAT1 | 214440_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker

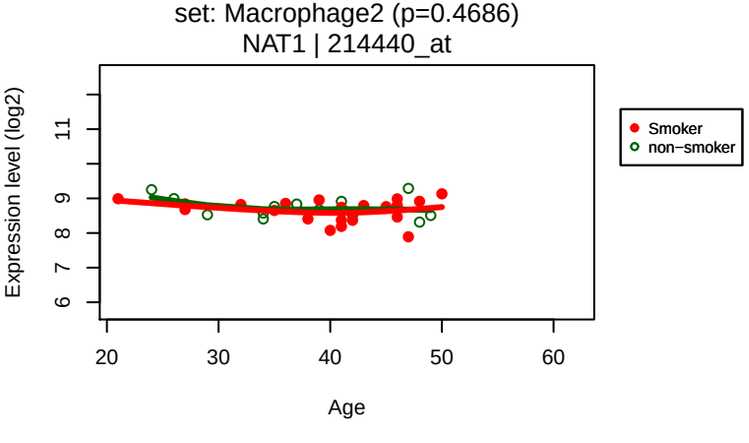

set: SmallAirway2 (p=0.8713)
NAT1 | 214440_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker

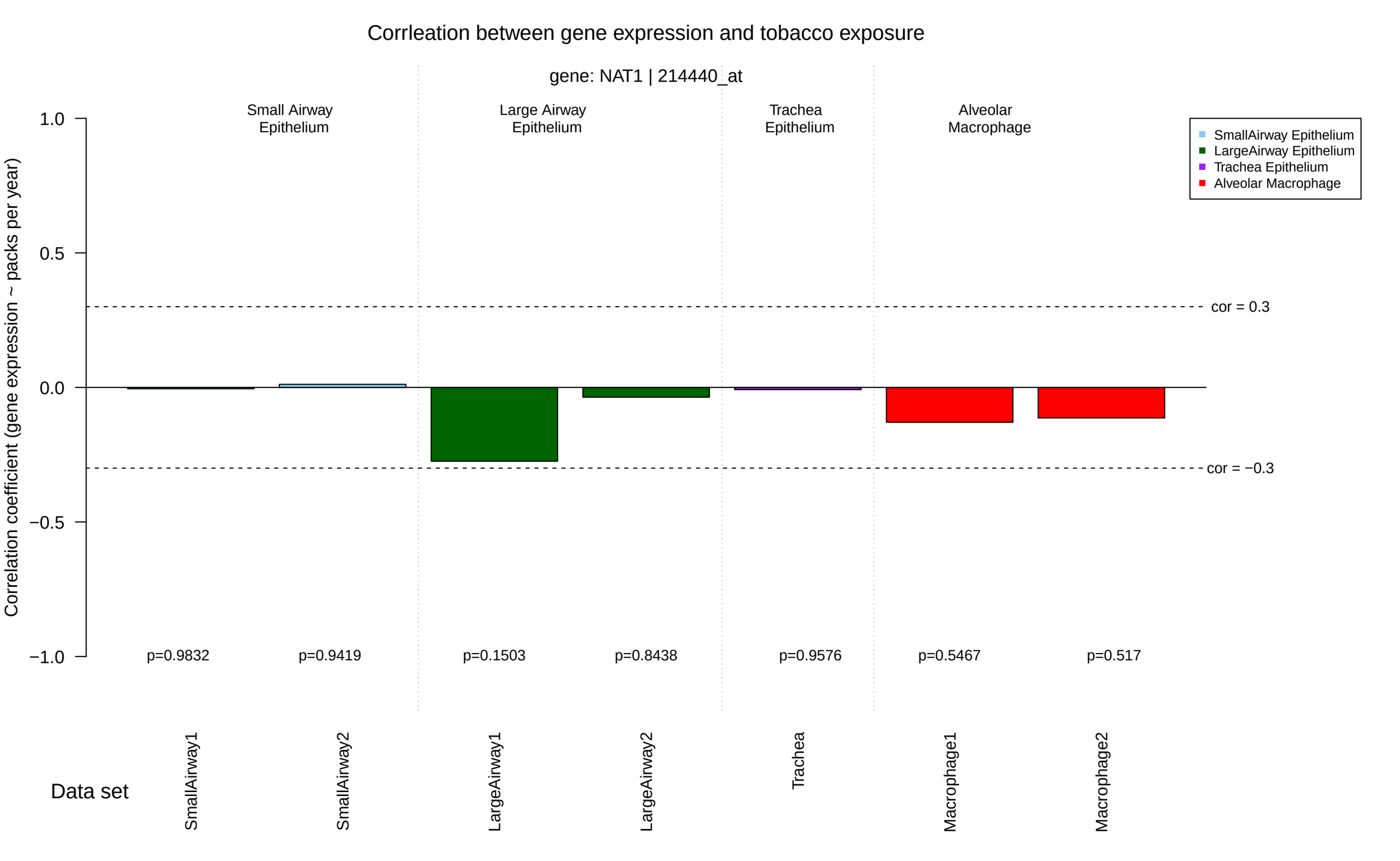

set: LargeAirway1 (p=0.017)
NAT1 | 214440_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1625)
NAT1 | 214440_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.8732)
NAT1 | 214440_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1923)
NAT1 | 214440_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4686)
NAT1 | 214440_at
Expression level (log2)
6
7
8
9
11
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NAT1 | 214440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9832
p=0.9419
p=0.1503
p=0.8438
p=0.9576
p=0.5467
p=0.517
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2